Student Name: _____

Student ID# _____

	Check which tutorial for pick-up
T01 Tuesday afternoons	
T02 Friday afternoons	

CHEMICAL ENGINEERING 2O04
Midterm #2

Duration 120 minutes (7:00pm – 9:00pm) **D.R. Latulippe**
McMaster University Term Test **Tuesday April 2nd, 2024**

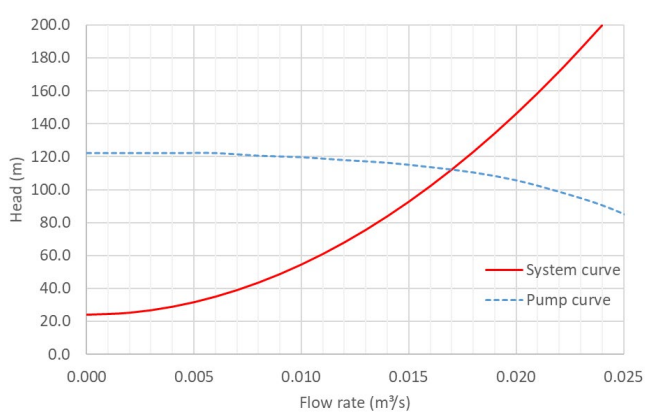
SPECIAL INSTRUCTIONS

1. The test includes 4 questions and 1 bonus question; ensure your copy of the test is complete.
2. Marks for each problem are given (**Total of 43 marks)**. Use this as a time-management guide; for example: $2\ marks\left(\frac{120\ minutes}{43\ marks}\right) \sim 6\ minutes$
3. The following materials are permitted during this examination:
   a) The Fox and McDonald's Introduction to Fluid Mechanics textbook (any edition and any version)
   b) Any printed notes or hand-written notes
   c) A tablet or computer in "airplane mode" with a document reader (e.g. Adobe Acrobat, Microsoft OneNote) open to review course material. **<u>Any student caught with a device not in airplane mode and/or with other applications opened will automatically receive a grade of 0 on the test.</u>**
   d) The McMaster Standard Calculator (Casio FX-991MS/MS+)
4. **The following materials are NOT permitted during this examination:**
   a) The internet.
   b) Any calculation software (e.g. Excel, MATLAB).
   c) Any applications to communicate with anyone.
5. Only tests written in pen will be considered if re-marking is requested (addition errors excepted).
6. **On page 11, we have given you a table with some useful conversion factors.**

For grading purposes only

Q1	Q2	Q3	Q4	Bonus	TOTAL

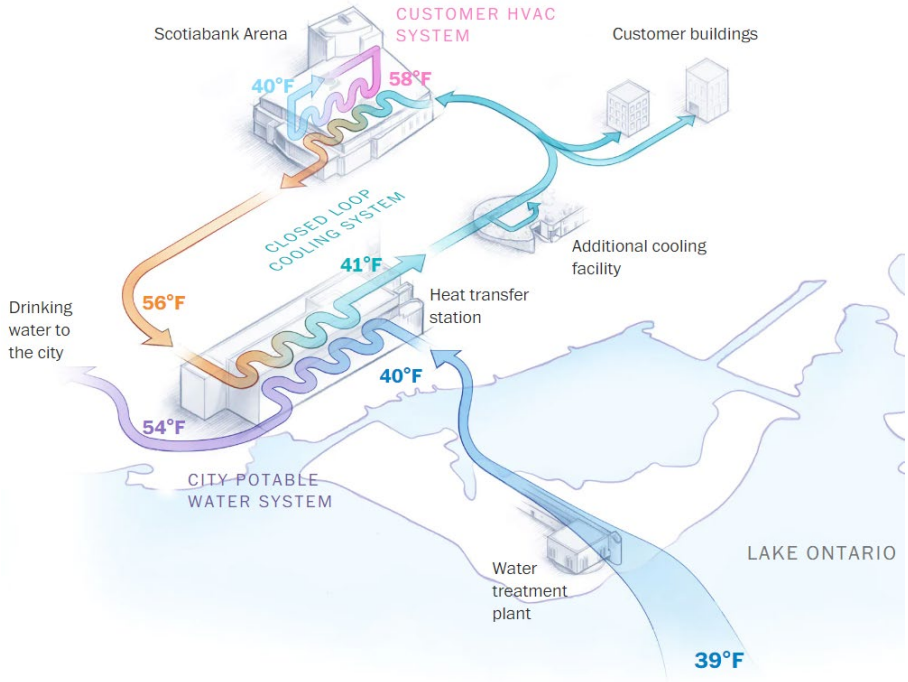
QUESTION 1 (5 MARKS) – circle the appropriate answer

For a turbulent flow field, which of the following statements is correct?	• Both the velocity and pressure are constant with respect to position. • The pressure is constant with respect to position but the velocity is not. • The velocity is constant with respect to position but the pressure is not. • Both the velocity and pressure are not constant with respect to position.
Your boss asks you to select a valve that minimizes pressure drop when completely open, to be installed in a particular piping system. Which of the following valves would you choose?	• Globe valve • Ball valve • Gate valve • Butterfly valve
Select the best response for why radial flow pumps normally designed with backward-curved blades (i.e. $\beta < 90°$).	• The pump will supply less energy if there is a surge in flow • The pump will supply more energy if there is a surge in flow • It minimizes the amount of fluid leakage. • It is easier to manufacture pumps that way.
What would be the operating point for the pump-piping system as described in the figure shown below?  	• 0.010 m^3/s • 0.014 m^3/s • 0.017 m^3/s • 0.020 m^3/s
Which statement correctly describes the change in NPSHA and NPSHR with increasing flow rate?	• NPSHA decreases and NPSHR increases • NPSHA increases and NPSHR decreases • NPSHA increases and NPSHR increases • NPSHA decreases and NPSHR decreases

QUESTION 2 – 10 MARKS

Toronto is home to the world's largest deep lake water cooling (DLWC) system – a rather innovative approach to cool many buildings in the downtown core (including the Scotiabank Arena) using cold water (4 ^{o}C = 39 ^{o}F) from the depths of Lake Ontario.

As shown in the accompanying figure, the process starts by drawing water from Lake Ontario into the Toronto Island water treatment plant via a network of large pipes. The water is treated for potable use and then pumped to a heat transfer station where it cools the incoming warm water from downtown buildings.



The pipe that brings water into the water treatment plant is 5 km long; due to the natural slope of the lake bottom, the inlet point to the pipe is at a depth of 83 meters (approximately one-third of the max depth of Lake Ontario) where the water is constantly at 4 ^{o}C.

a) Calculate the absolute pressure (kPa) at the lake depth of 83 meters.

NOTE: You can ignore the temperature variation of the lake water and use constant density value in your calculation.

(2 marks)

b) Calculate the pressure drop (kPa) across the 5 km pipe given the following:

- 127 million liters of cold water is drawn into the pipe each day
- The pipe has an internal diameter of 1.6 m, and is made of high-density polyethylene (roughness = 0.007 mm)

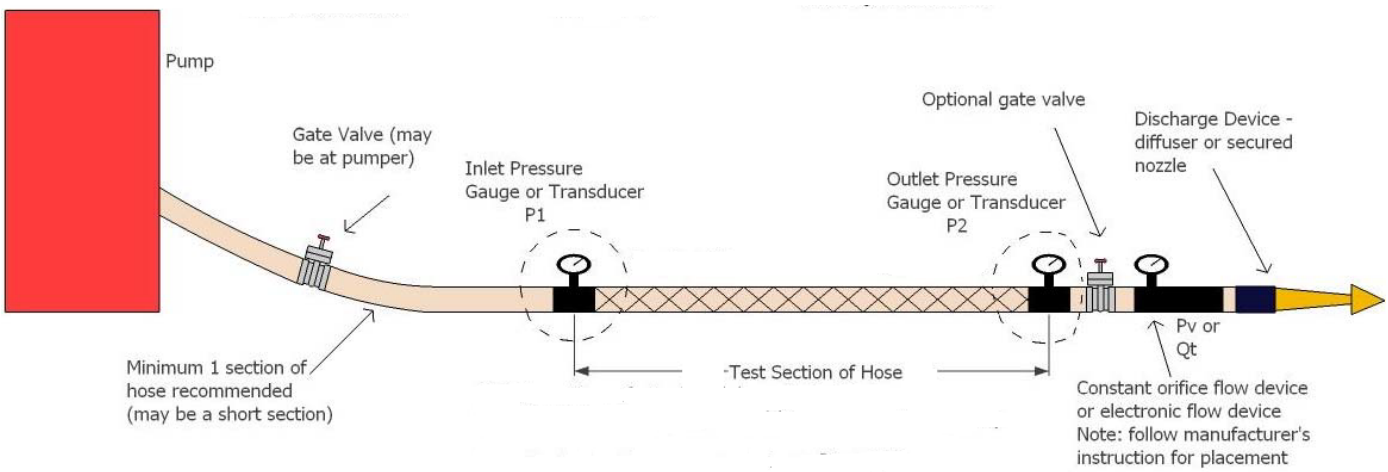
NOTE: You can ignore the contribution of any "minor losses".

(8 marks)

QUESTION 3 - 13 MARKS

One of the most common type of fire hoses (see picture to the right) is made of thermoplastic polyurethane (TPU) – these hoses are lighter and have better 'kink resistance' than the original hoses made from rubber.

a) Consider the scenario where a 50 ft length of 1.5" TPU fire hose (actual internal diameter = 1.6") is unrolled to lay flat on the ground) and connected to the test rig shown below. If the pressure drop along the 'test section of hose' is measured to be 40 psi (by two pressure measuring devices shown in figure) at a water flow rate of 160 gallons per minute (i.e. gpm), calculate the absolute roughness (mm) of the TPU fire hose. **(7 marks)**



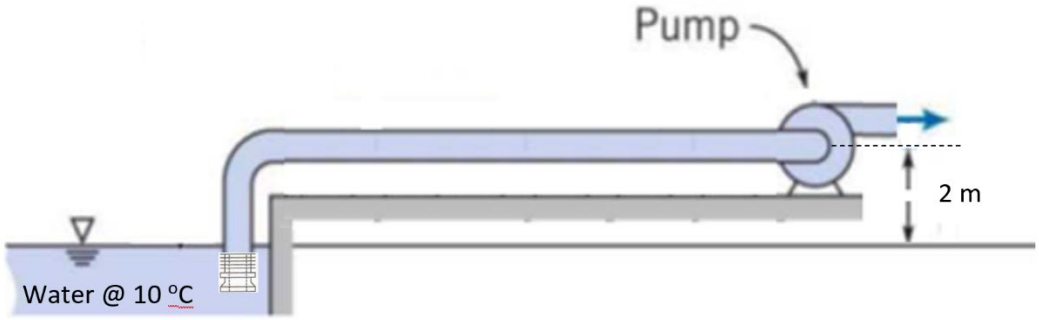
b) Using your result from part a), calculate the flow rate of water (m^3/s) for the same pressure drop reading (i.e. 40 psi) on a 100 ft length of 2.5” TPU hose (actual internal diameter = 2.7”) that is connected to the same fire hose test rig.

NOTE: If you are unsure how to calculate absolute roughness in part a), you can use the value 0.12 mm for part b).

(6 marks)

QUESTION 4 – 15 MARKS

Your colleague asks you to provide some consultation on the design of a system for pumping water (to be used as cooling water in an electric power plant located in Denver, Colorado) from a large open reservoir.



a) Calculate the maximum length (m) of the 3" schedule 40 galvanized iron pipe on the suction side of the FTI centrifugal pump DB10 (pump performance curve given on page 10 of the test) given the following:
   - The required volumetric flow rate of cooling water is 80 gallons per minute (i.e. gpm)
   - There is a basket strainer and 90° threaded regular elbow installed in the suction line

(9 marks)

b) You are told that the FTI centrifugal pump needs to be located 20 m beyond the answer in part a). Your colleague suggests a simple fix would be to swap the threaded elbow for a flanged elbow. Give a brief analysis of the suitability of your colleague's suggestion.

(2 marks)

c) As shown on the pump performance curve (on page 10 of the test), the FTI centrifugal pump DB10 can accommodate four different impellers:

#1 3.75" (95.2 mm)
#2 3.50" (88.9 mm)
#3 3.25" (82.6 mm)
#4 3.00" (76.2 mm)

Which impeller(s) would be suitable for the system as described in part a). Why?
If you identify more than one impeller, which one would you recommend be used? Why?

(4 marks)

BONUS QUESTION (6 MARKS)

At an elevation corresponding to sea-level, a medium-sized plastic cylindrical bottle (internal diameter = 10 cm) is mostly filled with 2L of water and then sealed shut – the volume of air in the bottle (above the water) is 0.00015 m^3. Then that bottle is put in a car and driven to the top of Mount Rainier (in Washington state), which is 4400 m above sea-level, where the atmospheric pressure is approximately 58 kPa.

a) Once you reach the top of Mount Rainier, you punch a small hole in the side of the sealed plastic bottle at a position that is 15 cm below the water level. What is the velocity (m/s) of the water exiting the bottle immediately after that small hole is made.

(3 marks)

b) Derive the correct expression for the velocity of the water through the punched hole as a function of the height of water above the punched hole.

NOTE: You can assume that the air trapped within the sealed bottle behaves as an ideal gas.

(3 marks)

Pump performance curve for Problem 4

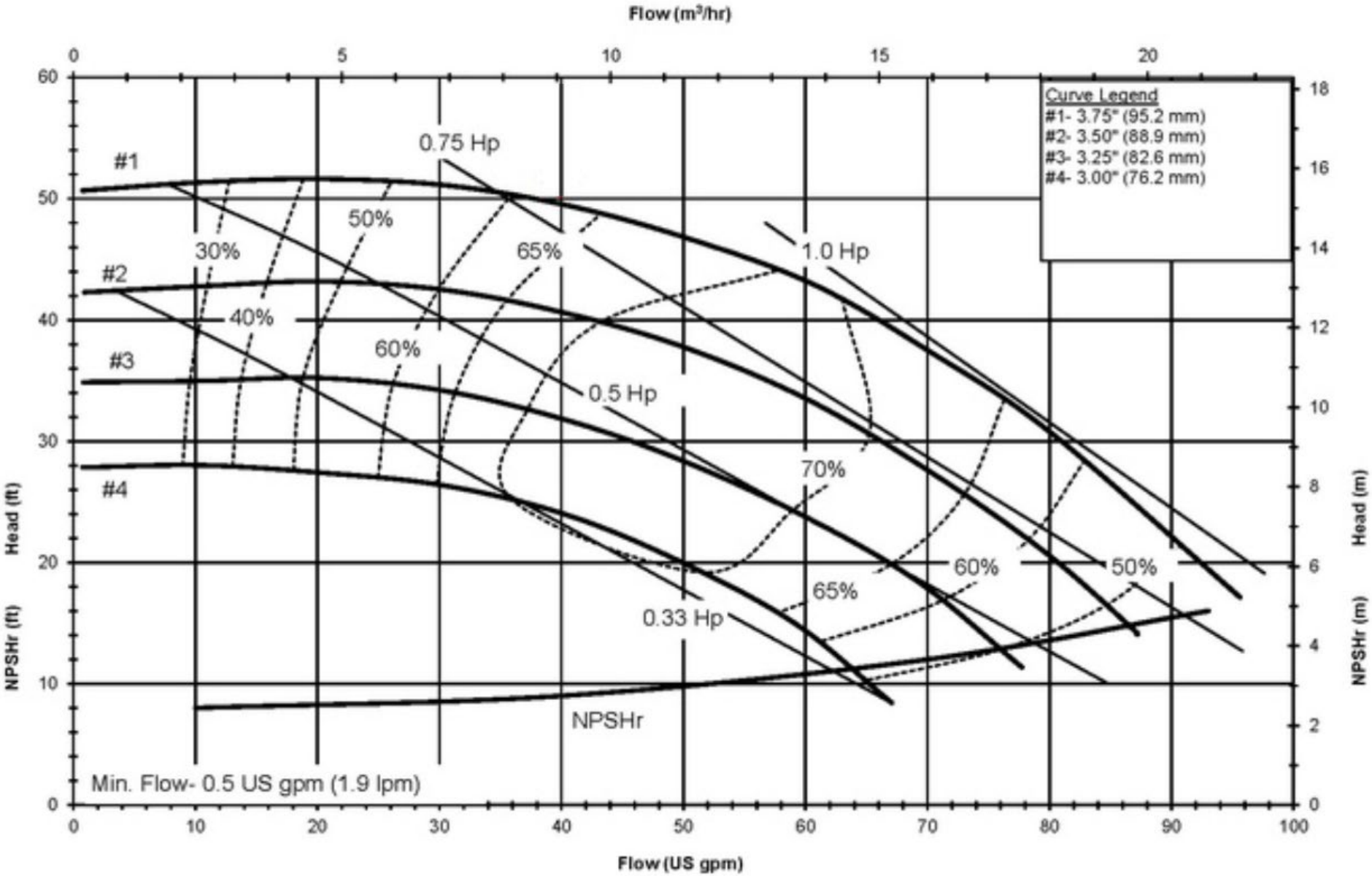


Table G.2

Conversion Factors and Definitions

Fundamental Dimension	English Unit	Exact SI Value	Approximate SI Value
Length	1 in.	0.0254 m	—
Mass	1 lbm	0.453 592 37 kg	0.454 kg
Temperature	1°F	5/9 K	—

Definitions:

Acceleration of gravity:	$g = 9.8066\ \text{m/s}^2$ ($= 32.174\ \text{ft/s}^2$)
Energy:	Btu (British thermal unit) ≡ amount of energy required to raise the temperature of 1 lbm of water 1°F (1 Btu = 778.2 ft · lbf)
	kilocalorie ≡ amount of energy required to raise the temperature of 1 kg of water 1 K(1 kcal = 4187 J)
Length:	1 mile = 5280 ft; 1 nautical mile = 6076.1 ft = 1852 m (exact)
Power:	1 horsepower ≡ 550 ft · lbf/s
Pressure:	1 bar ≡ 10^5 Pa
Temperature:	degree Fahrenheit, $T_F = \frac{9}{5}T_C + 32$ (where T_C is degrees Celsius)
	degree Rankine, $T_R = T_F + 459.67$
	Kelvin, $T_K = T_C + 273.15$ (exact)
Viscosity:	1 Poise ≡ 0.1 kg/(m · s)
	1 Stoke ≡ 0.0001 m^2/s
Volume:	1 gal ≡ 231 in.^3 (1 ft^3 = 7.48 gal)

Useful Conversion Factors:

Length:	1 ft = 0.3048 m	Power:	1 hp = 745.7 W
	1 in. = 25.4 mm		1 ft · lbf/s = 1.356 W
Mass:	1 lbm = 0.4536 kg		1 Btu/hr = 0.2931 W
	1 slug = 14.59 kg	Area	1 ft^2 = 0.0929 m^2
Force:	1 lbf = 4.448 N		1 acre = 4047 m^2
	1 kgf = 9.807 N	Volume:	1 ft^3 = 0.02832 m^3
Velocity:	1 ft/s = 0.3048 m/s		1 gal (US) = 0.003785 m^3
	1 ft/s = 15/22 mph		1 gal (US) = 3.785 L
	1 mph = 0.447 m/s	Volume flow rate:	1 ft^3/s = 0.02832 m^3/s
Pressure:	1 psi = 6.895 kPa		1 gpm = 6.309×10^{-5} m^3/s
	1 lbf/ft^2 = 47.88 Pa	Viscosity (dynamic)	1 $\text{lbf} \cdot \text{s/ft}^2$ = 47.88 $\text{N} \cdot \text{s/m}^2$
	1 atm = 101.3 kPa		1 $\text{g/(cm} \cdot \text{s)}$ = 0.1 $\text{N} \cdot \text{s/m}^2$
	1 atm = 14.7 psi		1 Poise = 0.1 $\text{N} \cdot \text{s/m}^2$
	1 in. Hg = 3.386 kPa	Viscosity (kinematic)	1 ft^2/s = 0.0929 m^2/s
	1 mm Hg = 133.3 Pa		1 Stoke = 0.0001 m^2/s
Energy:	1 Btu = 1.055 kJ		
	1 ft · lbf = 1.356 J		
	1 cal = 4.187 J		

THE END